Rain Detector Alarm using Transistors

School Science Project Report · Darlington pair (2× BC547) · No Arduino · No coding

Project	Rain Detector Alarm (Project 04)
Category	Electronics – working model
Power	9V battery (safe, low voltage)
Video	https://youtu.be/g0FPl0VcvvQ

1. Aim

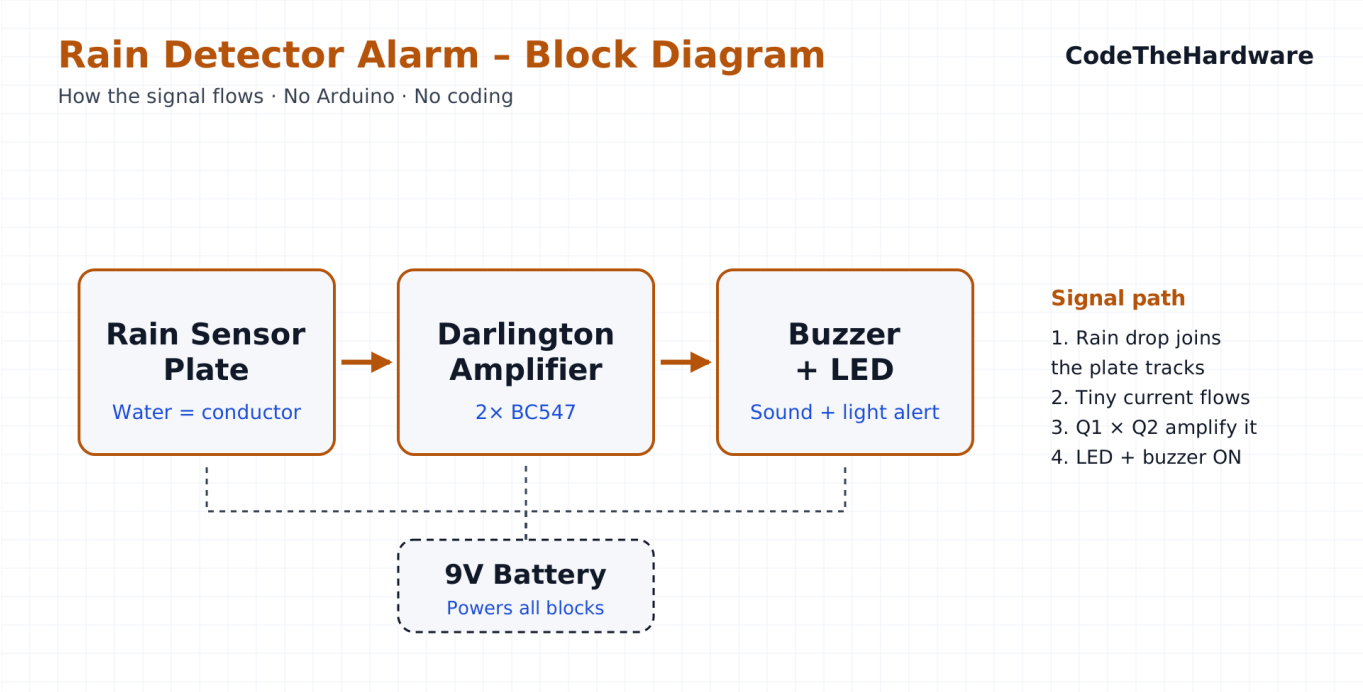
To build a simple electronic circuit that detects rain (water) and gives an alert using a buzzer and an LED, without any microcontroller or programming.

2. Materials Required

Component	Qty	Purpose	Approx. cost*
BC547 NPN transistor	2	Q1, Q2 – Darlington amplifier	₹4
1kΩ resistor	2	R1 base protection, R3 LED current limit	₹2
100k–470k resistor (optional)	1	R2 – stops false alarm when dry	₹1
Red LED	1	Light indication	₹2
Buzzer	1	Sound alert	₹20
Rain sensor plate	1	Detects water drops	₹40
9V battery + snap	1	Power supply	₹30
Breadboard + jumper wires	1 set	Connections	₹90

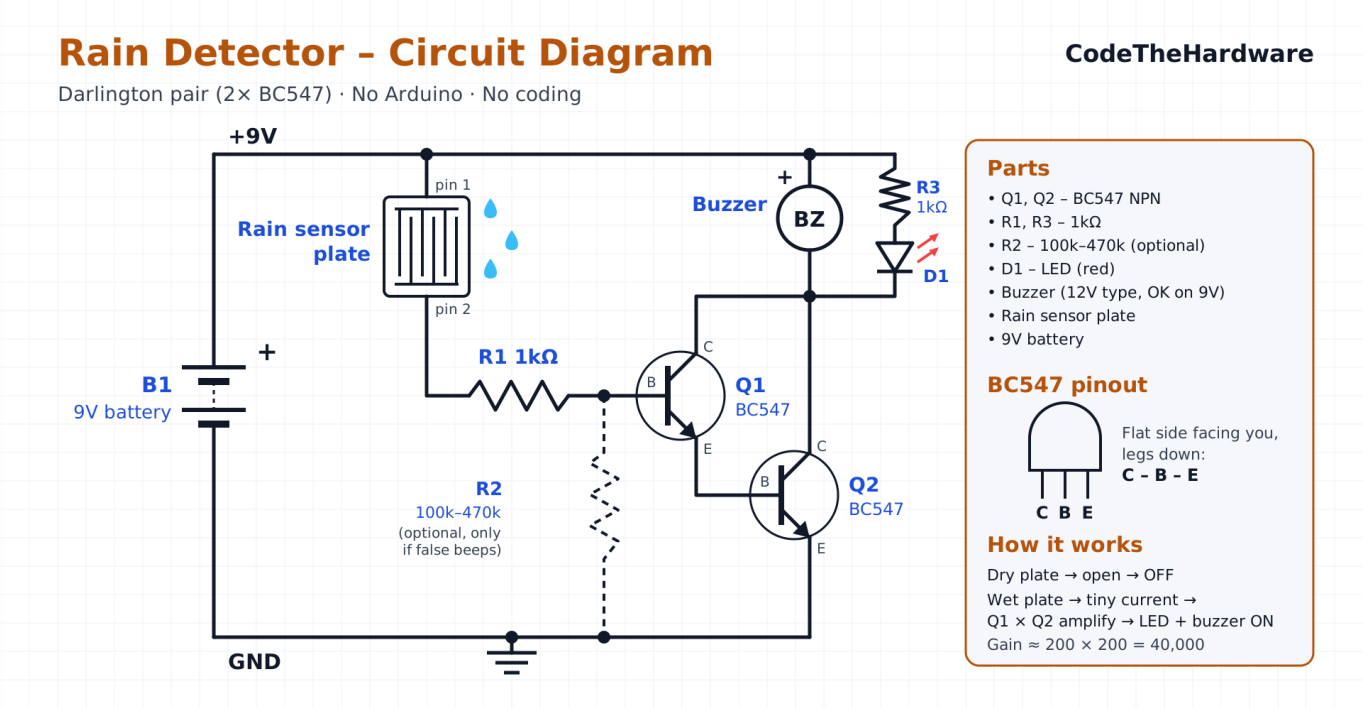
*Approximate prices; they vary by shop and city. Total about ₹190 (or about ₹100 if you already have a breadboard).

3. Block Diagram



PROJECT 04 · RAIN DETECTOR WITHOUT ARDUINO · NO CODING

4. Circuit Diagram



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Connections

From	To
Rain plate pin 1	+9V
Rain plate pin 2	1kΩ (R1) → Q1 Base
Q1 Emitter	Q2 Base
Q1 Collector + Q2 Collector	Buzzer (−) and LED (−)
Buzzer (+)	+9V
+9V	1kΩ (R3) → LED (+)
Q2 Emitter	GND (battery −)
R2 (optional)	Q1 Base → GND

BC547 pins: hold the flat side facing you, legs down → **C – B – E** (left to right).

5. Working Principle

- **Dry plate:** the two tracks of the rain plate are not connected, so no current flows into Q1's base. Both transistors are OFF, so the LED and buzzer are OFF. The circuit uses almost no battery in this state.
- **Wet plate:** rain water connects the tracks. Water is a weak conductor, so only a very small current (microamperes) flows through R1 into Q1's base.
- **Why 2 transistors?** One BC547 multiplies current by about 200. That is not enough for such a tiny current. In a **Darlington pair**, Q1's emitter current drives Q2's base, so the gains multiply: about 200 × 200 ≈ 40,000 (typical).
- **Example:** if the wet plate has 100kΩ, base current ≈ (9V − 1.4V) / 101kΩ ≈ 75 µA. Even a small part of the 40,000× gain is enough to fully switch ON Q2, so the buzzer (about 20–30 mA) and LED (about 7 mA) turn ON.
- When the plate dries, the path breaks again and the alarm stops automatically.

6. Procedure

1. Place both BC547 transistors on the breadboard, flat side facing you.
2. Connect the 9V battery to the breadboard rails (+ to red rail, − to blue rail).
3. Connect Q1 Emitter to Q2 Base, and join Q1 Collector with Q2 Collector.
4. Connect Q2 Emitter to GND.
5. Connect the buzzer: (+) to +9V, (−) to the joined collectors.
6. Connect +9V → 1kΩ (R3) → LED long leg; LED short leg to the joined collectors.
7. Connect rain plate pin 1 to +9V and pin 2 through 1kΩ (R1) to Q1 Base.
8. Test: keep the plate dry (no sound), then put water drops on it (LED + beep). Wipe it dry (alarm stops).
9. If it beeps when dry, add R2 (100k–470k) from Q1 Base to GND.

7. Observation Table

Measure the plate resistance with a multimeter (Ω range, e.g. 200k or 2M) and note the output. Fill in your own readings.

Condition of plate	Plate resistance (multimeter)	LED	Buzzer
Dry	Open / OL (very high)	OFF	OFF
Wet finger across tracks	_____ kΩ		
A few water drops	_____ kΩ		
Fully wet	_____ kΩ		
Wiped dry again	_____		

8. Result

The rain detector alarm was built successfully. When water falls on the rain sensor plate, the LED glows and the buzzer beeps. When the plate is dry, the alarm stays OFF.

9. Applications

- Alert to take dried clothes inside when it starts raining
- Water tank overflow alarm
- Floor water-leak alarm (near washing machine or water purifier)
- Rain alert for grains or crops kept outside to dry

10. Advantages and Limitations

Advantages	Limitations
Very low cost, easy to build	Plate corrodes over time because plain DC flows through the water (electrolysis)
No microcontroller or coding needed	Pure (distilled) water conducts very poorly and may not trigger it
Almost zero battery use while waiting	Only alerts; it cannot take any action on its own
Safe – works on 9V	The buzzer keeps beeping until the plate is dry

11. Industry Note

Real rain and water-level sensors avoid plain DC on the probes, because DC causes electrolysis and corrodes the tracks within weeks. They use stainless or gold-plated probes and switch the sensing voltage ON only for short moments (pulsed) or use AC excitation.

12. Future Scope

- Add a relay and a DC motor to automatically cover clothes (Arduino version)
- Use ESP32 to send a rain alert to a mobile phone
- Use a 555 timer to pulse the sensing voltage and reduce corrosion
- Add a small solar panel to charge the battery

13. Safety Precautions

- Use only a 9V battery. Never connect this circuit to 230V mains.
- Do not short the battery terminals.
- Keep water away from the battery and breadboard; only the rain plate should get wet.
- Take help from a parent or teacher while building.

14. Viva Questions and Answers

Q. What is the aim of this project?

A. To detect rain using a sensor plate and give a sound and light alert using transistors.

Q. Why does the circuit turn ON when water falls on the plate?

A. Water (with dissolved salts) conducts electricity, so it connects the two tracks and lets a small current flow into the transistor's base.

Q. Which transistor is used and what type is it?

A. BC547, an NPN bipolar junction transistor.

Q. What is a Darlington pair?

A. Two transistors connected so that the emitter of the first drives the base of the second. Their current gains multiply, giving very high gain.

Q. Why can't we use only one transistor?

A. The current through water is very small. One transistor's gain (about 200) is not enough to fully switch on the buzzer reliably.

Q. What is the use of the 1kΩ resistor at the base?

A. It limits the base current and protects Q1 if the plate gets fully shorted.

Q. Why is a resistor used with the LED?

A. To limit the current through the LED so it does not burn out.

Q. What happens if the plate is dry?

A. No base current flows, both transistors stay OFF, and the LED and buzzer are OFF.

Q. Does the circuit waste battery while waiting?

A. Almost no. Both transistors are OFF when the plate is dry, so very little current flows.

Q. What is the main limitation?

A. The plate slowly corrodes because DC flows through water (electrolysis).

Full build video: https://youtu.be/g0FPl0VcvvQ · Project help: codethehardware@gmail.com